Chapter 4

Capital Budgeting and Business Project Feasibility

BBA1005: Essential Finance for Entrepreneurs

MSME BUSINESS SCHOOL
ASSUMPTION UNIVERSITY

Learning Objectives

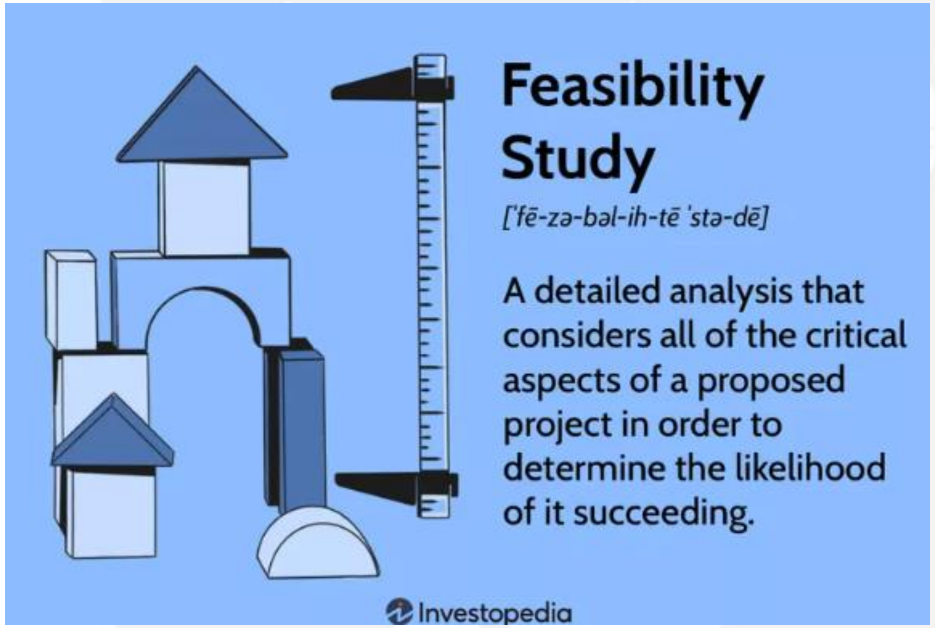
- To provide an understanding on the following concepts:
  - How to estimate project's revenues
  - How to estimate project's costs
  - How to estimate project's cash flow
- To be able to make a decision whether to accept or reject the project by applying the following criteria:
  - Net Present Value (NPV)
  - Internal Rate of Return (IRR)
  - Profitability Index (PI)
  - Payback period (PB)

Topics

1. Estimation of the Project's Revenues
2. Estimation of the Project's Costs
3. Estimation of the Project's Cash Flow
4. The Project Decision Criteria
   4.1. Net Present Value (NPV)
   4.2. Internal Rate of Return (IRR)
   4.3. Profitability Index (PI)
   4.4.Payback Period (PB)

Feasibility Study

- **Feasibility study** is an analysis of a proposed project to determine whether it has a potential to succeed.
- It includes various aspects of the project, i.e. economic, legal and regulations, technical, production, marketing, and **finance**.
- **Financial feasibility** focuses on the financial aspects of the project. It determines whether a proposed project can generate a sufficient return to pay for all costs.



https://www.investopedia.com/terms/f/feasibility-study.asp

Capital Budgeting

- Capital budgeting is the method of evaluating financial viability of a capital investment over the life of the investment.
- Capital investments are long-term investments in which the assets involve useful lives of multiple years.

Capital Budgeting

- Examples include
  - Constructing a new production facility
  - Investing in a new shop or starting a retail business
  - Purchasing or investing in machinery or equipment
  - Investing in an outside venture
  - Adding a new product line

Capital budgeting focuses on **cash flow**, rather than account profit.

Using Capital Budgeting to Assess Feasibility of a Business Project

Steps to assess feasibility of a business project include

1. Estimation of the project's revenues.
2. Estimation of the project's costs.
   (i.e. initial costs, fixed cost, variable costs, opportunity costs)
3. Estimation of the project's cash flow
4. Calculate the project's decision criteria
   - Net present value (NPV)
   - Internal rate of return (IRR)
   - Profitability index (PI)
   - Payback period (PB)
5. Making a decision to accept or reject the project.

Project’s Revenues

Estimation of Project's Revenue

- **Revenue projection** involves an estimation of how much money a company will generate over the project's life.
- Estimation of revenue involves **two** major components

1. Price
2. Quantity Sales

Sales revenue = Price × Quantity Sales

https://www.indeed.com/career-advice/career-development/revenue-projections

Estimation of Project's Revenue

- **Steps to estimate total revenues of the project**

1. Setting a **price** of a product (or service).
2. Estimate **quantity** sales.
3. Estimate a percentage **growth in price** over the project's life.
4. Estimate a percentage **growth in quantity** sales over the project's life.
5. Calculate the project's revenue in each year.

Sales Revenue = Price × Quantity Sales

1. Setting Price of Product (or service)

Q: How to **set price** of a product (or service)?

A: There are various methods to price a product (or service).

The most common pricing strategies include:

1. **Cost-plus pricing**
2. **Competitive pricing**
3. **Price skimming**
4. **Penetration pricing**
5. **Value-based pricing**

https://www.bdc.ca/en/articles-tools/marketing-sales-export/marketing/pricing-5-common-strategies

1. Setting Price of Product (or service)

1. Cost-plus pricing

- It is the most common use for **start-up** and **small business.**
- It involves the calculation of your costs and then add a mark-up on your cost.
- It can be calculated by

Price = Cost × (1 + Mark-up)

- Mr. A buys a T-shirt for 100 baht and he would like to have a profit margin of 30%; thus, he should sell at the price of **130** baht

Price = 100 × (1 + 0.3)

https://www.patriotsoftware.com/blog/accounting/cost-plus-pricing-strategy/
https://www.bdc.ca/en/articles-tools/marketing-sales-export/marketing/pricing-5-common-strategies

Cost Plus Pricing

Cost Plus Pricing Pros and Cons

Pros:

1. Simple to use
2. Predictably covers costs
3. Easy to justify

Cons:

1. Inefficient
2. Ignores market conditions
3. Doesn't show value

https://www.google.com/search?q=cost+plus+pricing+formula&source=lnms&tbm=isch&sa=X&ved=2ahUKEwjdq_rvkYf8AhUn4XMBHUF-BJEQ_AUoAXoECAEQAw&biw=1366&bih=656&dpr=1#imgrc=omCR5JXUeaTe9M

1. Setting Price of Product (or service)

2. Competitive pricing

- Set a price based on your **competitor's price**, rather than your cost.
- Set the price **equal** to competitors' price.
- Often use for the product that has been sold in the market for a while (or for a **long time**) and there are **many substitute** products.
- Often use when a company sells **similar** product to competitors.
- **Simple** and can **quickly** respond to the market changes.
- **Disadvantage**: customers are **not** aware of or value your product

https://www.bdc.ca/en/articles-tools/marketing-sales-export/marketing/pricing-5-common-strategies

1. Setting Price of Product (or service)

3. Price skimming

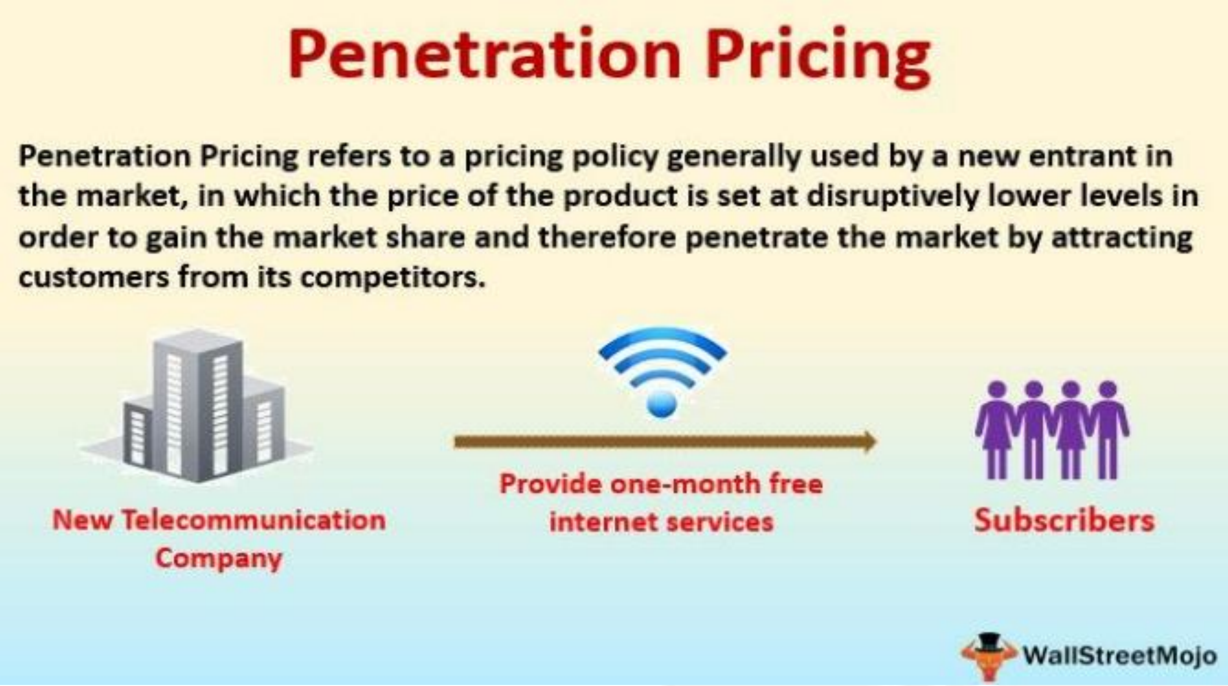
- Set a **high price** for a new product and **lower the price later** as more competitors entering the market.
- It generates a high short-term profit.
- It can be used with an **innovative** product in order to attract **insensitive** price customers who want to try new product. (ex. New iPhone from Apple, or Nike products)

https://www.coursera.org/articles/skim-pricing
https://www.investopedia.com/terms/p/penetration-pricing.asp

1. Setting Price of Product (or service)

4. Penetration pricing.

- Set a **low price** for a new product to enter a competitive market and build customers' base, then **raise the price later**.
- It uses to attract a wide rage of customers at first to build a market share (Ex. Memberships, Internet providers, Shopee, lineman)

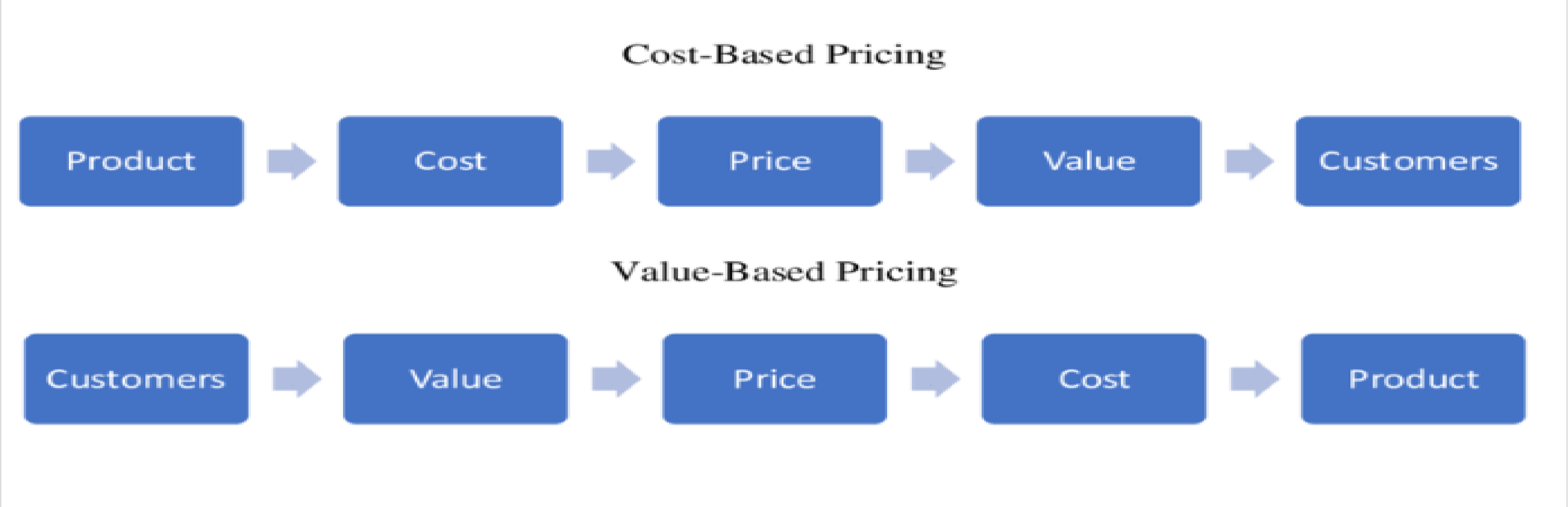


https://www.coursera.org/articles/skim-pricing
https://www.investopedia.com/terms/p/penetration-pricing.asp
https://www.wallstreetmojo.com/penetration-pricing/

1. Setting Price of Product (or service)

5. Value-based pricing

- Set a price based on **customers' perception on value** of a product.
- Charge customers at the price that customers are willing to pay.
- It is complex and requires a significant market research (i.e. customers survey)
- Usually use for **unique** and **highly valuable** products.



https://www.researchgate.net/figure/Cost-Based-vs-Value-Based-Pricing_fig1_343575442

2. Estimate Quantity sales

2. **Quantity sales** – The number of units of output that a company is expects to sell. It is quantity of goods or services that buyers are willing and able to buy which is determined by several factors such as:
   - Production capacity
   - Economic conditions (growth and income)
   - Number of population (e.g. baby product)
   - Seasonal
   - Advertising
   - Number of competitors

Note: Marketing survey or research may be used to help estimate quantity sales

3. Growth in Price

- To estimate the project's revenue, it may be appropriate to assume that the price of a product (or service) grows over years.
- **Price** may increase due to a significant **inflation**.
- Years gone by, inflation results in a significant increase in costs of raw materials, labors, and overhead (i.e. utilities).
- (Note: It may appropriate to assume that the price increase by **average inflation** rate).

4. Growth Quantity Sales

- To estimate the project's revenue, it may be appropriate to assume that quantity sales grow over years.
- **Quantity sales** may increase since at the beginning customers' awareness of the product is still low. Once the product is introduced for years, **awareness and purchase** may increase; thus, quantity sales increase.

5. Calculate Project's Revenue

Input Table	
Price	$11.6
% Growth in price after the first year	2%
Quantity Sales	550,000
% Growth in quantity after the first year	4%

- **Should SP Baker Co.Ltd. purchase and install this new machine?**
- If SP Baker Co. Ltd. has installed a new baking machine, this machine is expected to use for 5-years and can be sold after that.
- The estimated price of its bakery product and sales are available in the input picture.
- This input is used to calculate the project's revenue over the 5-year project.

5. Calculate Project's Revenue

Year	0	1	2	3	4	5
Price (2% growth)		11.6	11.83	12.07	12.31	12.56
Quantity Sales (4% growth)		550000	572000	594880	618675	643422
Total Revenue		6380000	6766760	7180201.6	7615889.25	8081380.3

Initial Price is $11.6 (in year 1)

- Sales Prices are expected to grow at 2%; thus,
- In year 2, price is $11.60 × (1 + **0.02**) = $**11.83**
- In year 3, price is $11.83 × (1 + 0.02) = $**12.07**
- In year 4, price is $12.07 × (1 + 0.02) = $**12.31**
- In year 5, price is $12.31 × (1 + 0.02) = $**12.56**

5. Calculate Project's Revenue

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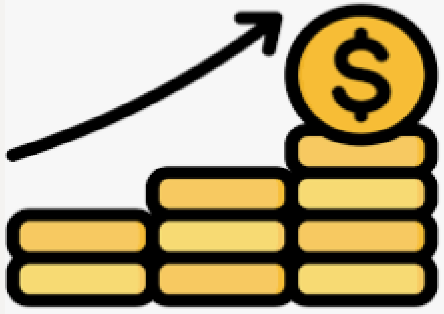
Initial Sales is 550,000 (in year 1)

- Sales are expected to grow at 4%; thus,
- In year 2, quantity sales is 550,000 × (1 + **0.04**) = **572,000**
- In year 3, quantity sales is 572,000 × (1 + 0.04) = **594,880**
- In year 4, quantity sales is 594,880 × (1 + 0.04) = **618,675**
- In year 5, quantity sales is 618,675 × (1 + 0.04) = **643,422**



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Total Revenue = Price × Quantity Sales

Sale Revenue = P * Q

Y1 => 11.6 × 550,000 = **$6,380,000**

Y2 => 11.83 × 572,000 = **$6,766,760**

Y3 => 12.07 × 594,880 = **$6,766,760**

Y4 => 12.31 × 618,675 = **$7,615,889.25**

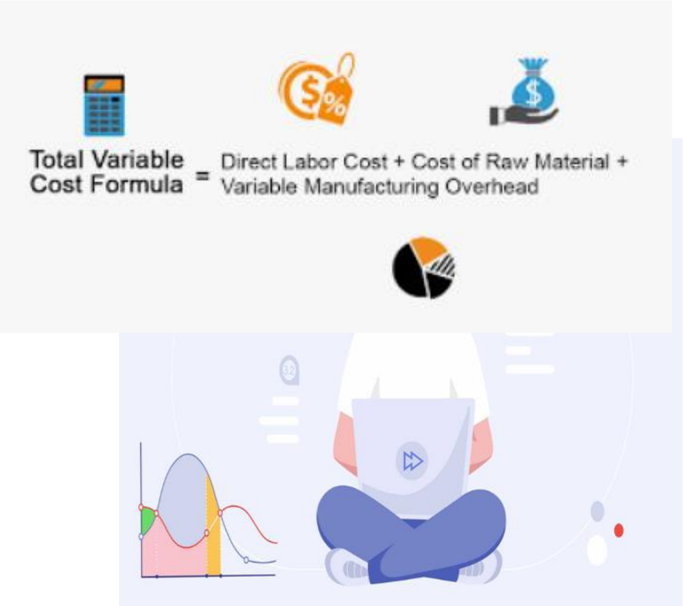
Y5 => 12.56 × 643,422 = **$8,081,380.3**

Project's Costs

Estimation of Project's Costs

There are various types of costs that we should know to estimate project's costs which are

1. Initial Costs
2. Fixed Costs
3. Variable Costs
4. Opportunity Costs



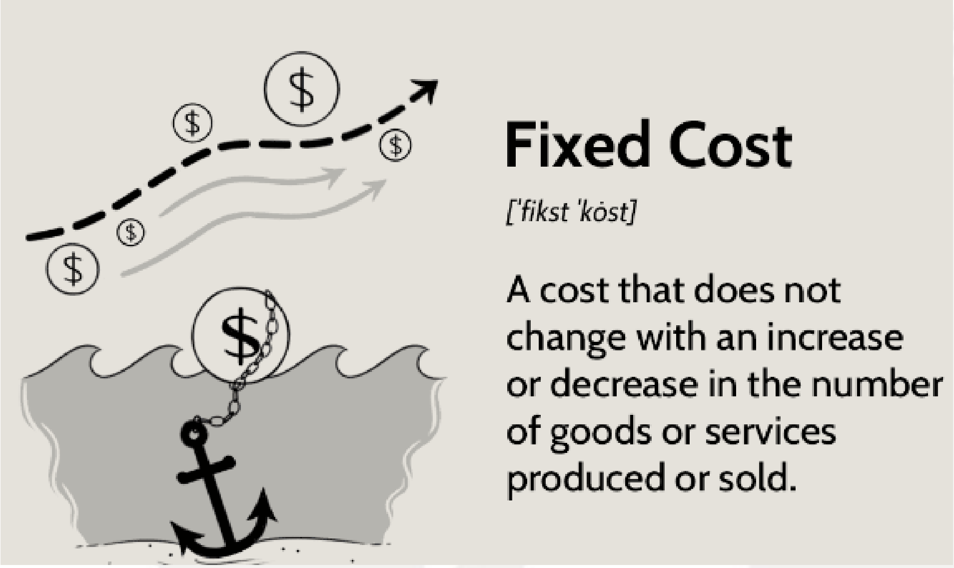
Estimation of Project's Costs

1. **Initial Costs** (**year 0**)
   - Costs (or money) that a firm must pay before starting to sell a product.
   - Capital construction
   - Installation of machine and equipment
   - Renovation of building and facilities

Estimation of Project's Costs

2. Fixed Costs

- Costs that do **not** increase or decrease with the number of goods or services sold.
- **Rent** of office building or shop
- **Lease** of vehicles or machine
- Utility Bills (i.e. electricity, water, internet)
- **Salaries**

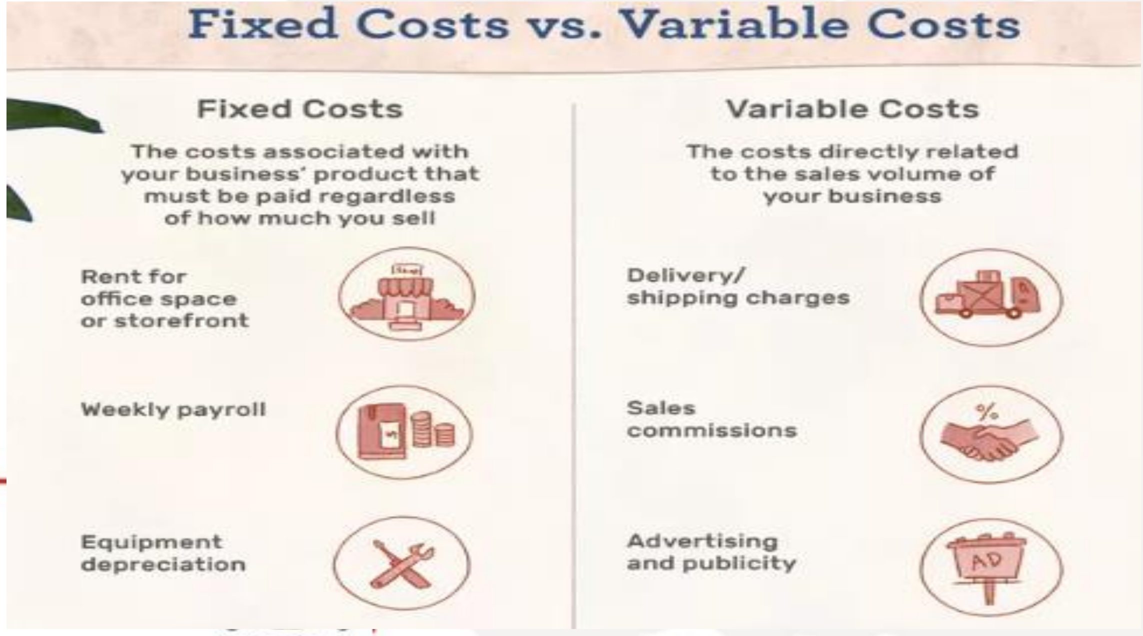


https://www.google.com/search?q=Fixed+Cost&tbm=isch&ved=2ahUKEwihscKEy4f8AhV1KbcAHXtnARkQ2-cCegQIABAA&oq=Fixed+Cost&gs_lcp=CgNpbWcQAzIECAAQQzIECAAQQzIECAAQQzIECAAQQzIECAAQQzIECAAQQzIECAAQQzIFCAAQgAQyBQgAEIAEMgUIABCABDoICAAQgAQQsQNQ8wZY-iFgySNoAHAAeAKAAWyIAYEKkgEEMTIuMpgBAKABAaoBC2d3cy13aXotaW1nsAEAwAEB&sclient=img&ei=flihY6H6H_XS3LUP-86FyAE&bih=656&biw=1366#imgrc=ewalvM5QSSBYyM

Estimation of Project's Costs

3. Variable Costs

- Costs that change as the number of goods or services sold increases or decreases.
- **Direct Materials** - The raw materials that go into the production
- **Sales Commissions** – Payment to employees that is based purely on the sales they make
- **Piece-rate labor**
- **Packaging costs**



https://www.thebalancemoney.com/a-guide-to-fixed-and-variable-costs-of-doing-business-393479

Estimation of Project's Costs

4. Opportunity Costs

- Costs of using the firm's owner **self-owned** resources
- Ex. Company A owns land at a current market value of 5 Million baht which can be rented at 25,000 baht per month (or 300,000 baht a year).
- If company A uses this land to implement the project, the company must forgo the rent of 25,000 baht a month.
- **Forgoing the rent of $25,000 (or 300,000 baht a year)** is the opportunity cost of implementing the project.
- Opportunity cost must be included as the project's costs.

Estimation of Project's Costs

Steps to estimate costs of the project

1. Collect data on initial costs
2. Estimate fixed costs, variable costs, and opportunity costs
3. Estimate growth in fixed costs, variable costs per units, and/or opportunity cost
4. Calculate the project's costs in each year

Total Variable Costs = Variable Cost per Unit × Quantity Sales

Calculate Project's Costs

Input Table	
New Baking Machine (**year 0**)	**3,400,000**
Salvage value (after 5 years)	**400,000**
Fixed costs	**2,000,000**
Growth in Fixed Costs	**2%**
Variable costs	**$6 per unit**
Growth in variable costs	**2%**
Opportunity Costs	**0**

- SP Baker co., ltd. has installed a new baking machine.
- This machine is expected to be used for 5 years and it can be sold after that.
- Estimated price of its bakery product and sales are available in the input picture.
- This input is used to calculate the project's costs over the 5-year project.

Calculate Project's Costs

Year	0	1	2	3	4	5
	3,400,000					
Fixed Costs (2% growth)		2,000,000	2,040,000	2,080,800	2,122,416	2,164,864.32
Variable Costs per unit (2% growth)		6	6.12	6.24	6.37	6.49
Quantity Sales (4% growth)		550000	572000	594880	618675	643422
Total Variable Costs (VC × Q)		3300000	3500640	3712051.2	3940959.75	4175808.78

Project's Cash Flow

Project's Cash Flow

Important Issues on the Project's Cash Flow

1. **Use cash flow** (not accounting profit)
   - When making the project's decision, analysis should be based on **cash flow** from cash flow statement, not accounting profit from income statement
   - Depreciation which is the non-cash expense should be excluded from cash flow (unless it is used to calculate tax)

2. **Opportunity Costs**
   - If there is an opportunity cost, opportunity costs must be added to the project's costs

Project's Cash Flow

Important Issues on the Project's Cash Flow

3. Effects of Depreciation

- Depreciation expenses must be deducted throughout the life of the project's assets in order to calculate tax.
- Depreciation shelters a firm from tax.
- However, depreciation is not a cash outflow.
- Depreciation must be added back when estimating the project's cash flow

4. Interest expenses

- When estimating the project's cash flow, interest expenses are ignored. (Interest is used to calculate NPV later.)

Steps to Estimate Project's Cash Flow

Year	0	1	2	3	4	5
Initial Cost (building &machine)						
Initial Opportunity Costs						
Sales Revenue						
- Fixed Costs						
- Total Variable Costs						
- Depreciation						
Earning before interest and tax (EBIT)						
Tax (20%)						
Net operating profit after tax						
+ Add back depreciation						
- Opportunity Costs						
+ Salvage value (if any)						
- Tax on salvage value						
Project's Cash Flow						

Steps to Estimate Project's Cash Flow

Year	0	1	2	3	4	5
Initial Cost (building &machine)	-3,400,000					
Initial Opportunity Costs	0					
Sales Revenue						
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Earning before interest and tax (EBIT)						
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Growth in variable costs	**2%**
Opportunity Costs	**0**

Project’s Cash Flow

Year	0	1	2	3	4	5
Initial Costs (Building & Equipment)	-3,400,000					
Initial Opportunity costs	0					
Total Revenue		6380000	6766760	7180201.6	7615889.25	8081380.32
- Fixed Costs		2000000	2040000	2080800	2122416	2164864.32
- Total Variable Costs		3300000	3500640	3712051.2	3940959.75	4175808.78
- Depreciation (straight line)		600,000	600,000	600,000	600,000	600,000
EBIT		480,000	626,120	787,350	952,514	1,140,707
- Tax (20%)		96000	125224	157470.08	190502.7	228141.444
Net Operating Profit After Tax		384,000	500,896	629,880	762,011	912,566
Add back depreciation		600,000	600,000	600,000	600,000	600,000
- Opportunity costs		0	0	0	0	0
Plus salvage value		0	0	0	0	400,000
- Tax on salvage value (20%)		0	0	0	0	80000
Project's Cash Flow	-3,400,000	984,000	1,100,896	1,229,880	1,362,011	1,832,566

Calculate Project's Revenue

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Income Statement

- **Depreciation Expenses** – a decrease in asset's value over time due to wear and tear
- Depreciation is a **non-cash** expense.
- Calculation is based on the used up of **long-term assets** (e.g., plant, machine, furniture, and motor vehicles)
- **Depreciation expenses**

$$= \frac{\text{(Value of Asset – Residual Value)}}{\text{Useful life of asset}}$$

- Residual value is the amount expected to realize at the end of its useful life
- Useful life is the expected life on an asset which an organization can derive benefits from it.

Depreciation

- Simple method to calculate depreciation
- **Depreciation = (Value of Asset – Residual Value) / Useful life of asset = (3,400,000 - 400,000) / 5**

$$\text{Depreciation} = \frac{\text{Value of Asset} - \text{Residual Value}}{\text{Useful life of asset}} = \frac{3{,}400{,}000 - 400{,}000}{5}$$

= 600,000 each year

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- Tax on salvage value (20%)		0	0	0	0	80000
Project's Cash Flow	-3,400,000	984,000	1,100,896	1,229,880	1,362,011	1,832,566

Project's Cash Flow

Year	0	1	2	3	4	5
Initial Costs (Building & Equipment)	-3,400,000					
Initial Opportunity costs	0					
Total Revenue		6380000	6766760	7180201.6	7615889.25	8081380.32
- Fixed Costs		2000000	2040000	2080800	2122416	2164864.32
- Total Variable Costs		3300000	3500640	3712051.2	3940959.75	4175808.78
- Depreciation (straight line)		600,000	600,000	600,000	600,000	600,000
EBIT		480,000	626,120	787,350	952,514	1,140,707
- Tax (**20% of EBIT**)		96000	125224	157470.08	190502.7	228141.444
Net Operating Profit After Tax (NOPAT)		384,000	500,896	629,880	762,011	912,566
Add back depreciation		600,000	600,000	600,000	600,000	600,000
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Project’s Decision Criteria

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Project's Decision Criteria

1. **Net Present Value (NPV)**
2. **Internal Rate of Return (IRR)**
3. **Profitability Index (PI)**
4. **Payback period (PB)**

Net Present Value (NPV)

Net Present Value

Net Present value

- The present value of a project's cash **inflow minus** the present value of its **costs.**
- The value that directly contributes to a corporation and shareholders' wealth.
- It indicates **"the value ($) that an entrepreneur will receive if implementing the project".**
- **Best Screening Criterion**
- **Accept the project when NPV is positive**
- The **larger the NPV**, the **more value ($)** are added to an entrepreneur.

Step to find Net Present Value

1. Calculate the present value of each cash flow discounted at the project's risk-adjusted cost of capital.
2. Sum the discounted cash flow to find the project NPV

The equation to find NPV is as follow:

$$\mathrm{NPV} = \mathrm{CF}_0 + \frac{CF_1}{(1+r)^1} + \frac{CF_2}{(1+r)^2} + \ldots + + \frac{CF_N}{(1+r)^N}$$

Where CF_t is the expected cash flow at time t

r is the project's **cost of capital** (or WACC)

(Ex. Interest on bank loan)

N is the life-time of the project

Example: Project's Cash Flow

Year	0	1	2	3	4	5
Initial Costs (Building & Equipment)	-3,400,000					
Initial Opportunity costs	0					
Total Revenue		6380000	6766760	7180201.6	7615889.25	8081380.32
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Plus salvage value		0	0	0	0	400,000
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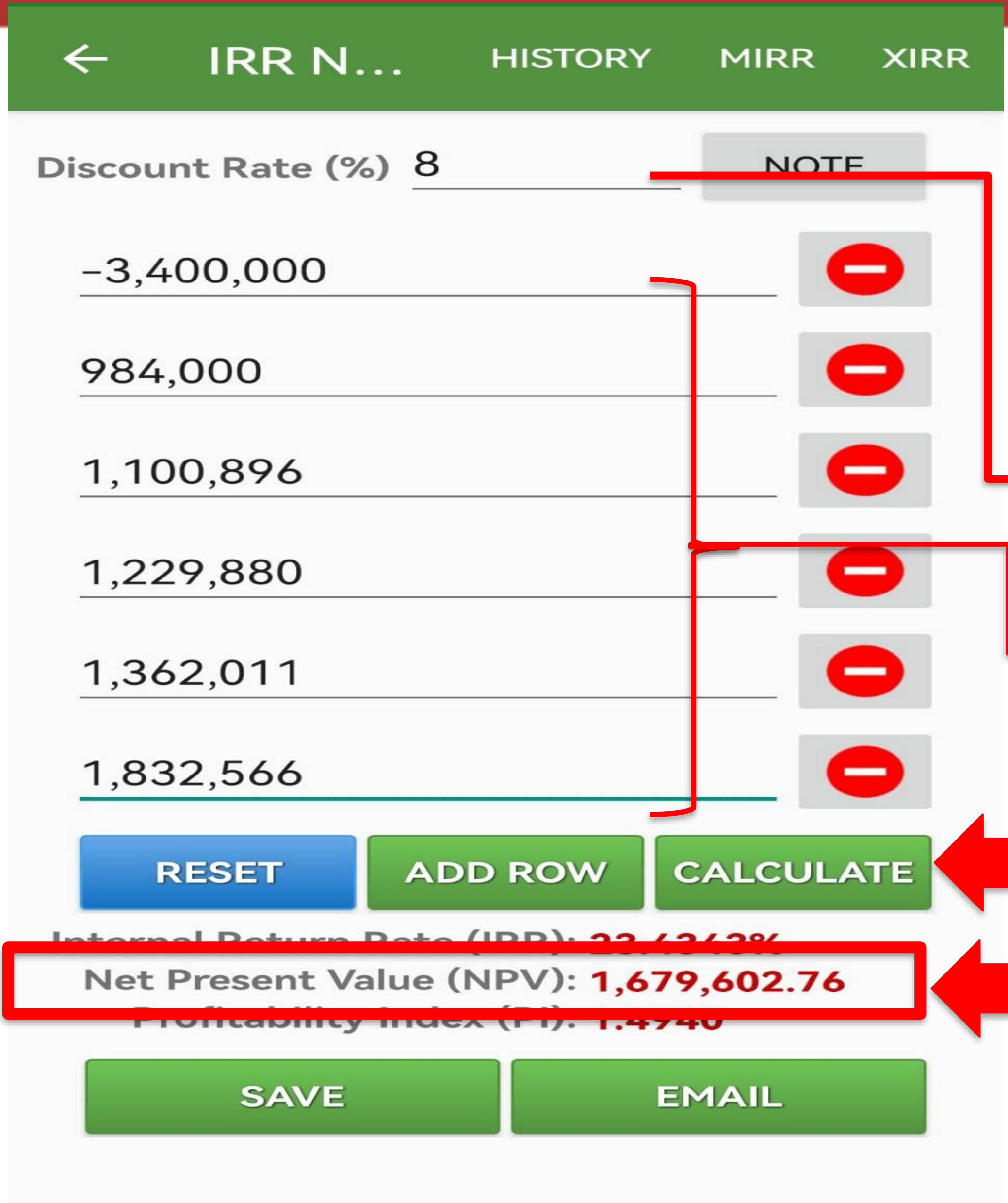
Project's Cash Flow	**-3,400,000**	**984,000**	**1,100,896**	**1,229,880**	**1,362,011**	**1,832,566**

Net Present Value (NPV): Calculator



- **Use IRR NPV calculator**
- Put a discount rate – **cost of capital** used to implement the project (Ex. If an entrepreneur finances this project by a bank loan, the cost of capital is the **interest rate** that an entrepreneur must payback to the bank.)
- Assume that an interest rate on loan is **8%** per year.

Put cash flow that estimated earlier in these lines

Press Calculate

Answer: NPV = 1,679,602.76
If implementing the project, it will generate cash flow to an entrepreneur by 1,679,602.76 baht.
Accept the project since NPV is positive.
The higher NPV, the better

Net Present Value (NPV): Excel

For Excel, use **NPV** function

=NPV(rate, value1, Value2, …….) + initial cash flow (year 0)

=NPV(8%,C2:G2) + B2

B4 =NPV(8%,C2:G2)+B2

	A	B	C	D	E	F	G
1	Year	0	1	2	3	4	5
2	Project's Cash Flow	(3,400,000.00)	984,000.00	1,100,896.00	1,229,880.00	1,362,010.00	1,832,565.00
3							
4	Net Present value	1,679,601.34					
5							

Internal Rate of Return (IRR)

Internal Rate of Return (IRR)

Internal Rate of Return (IRR)

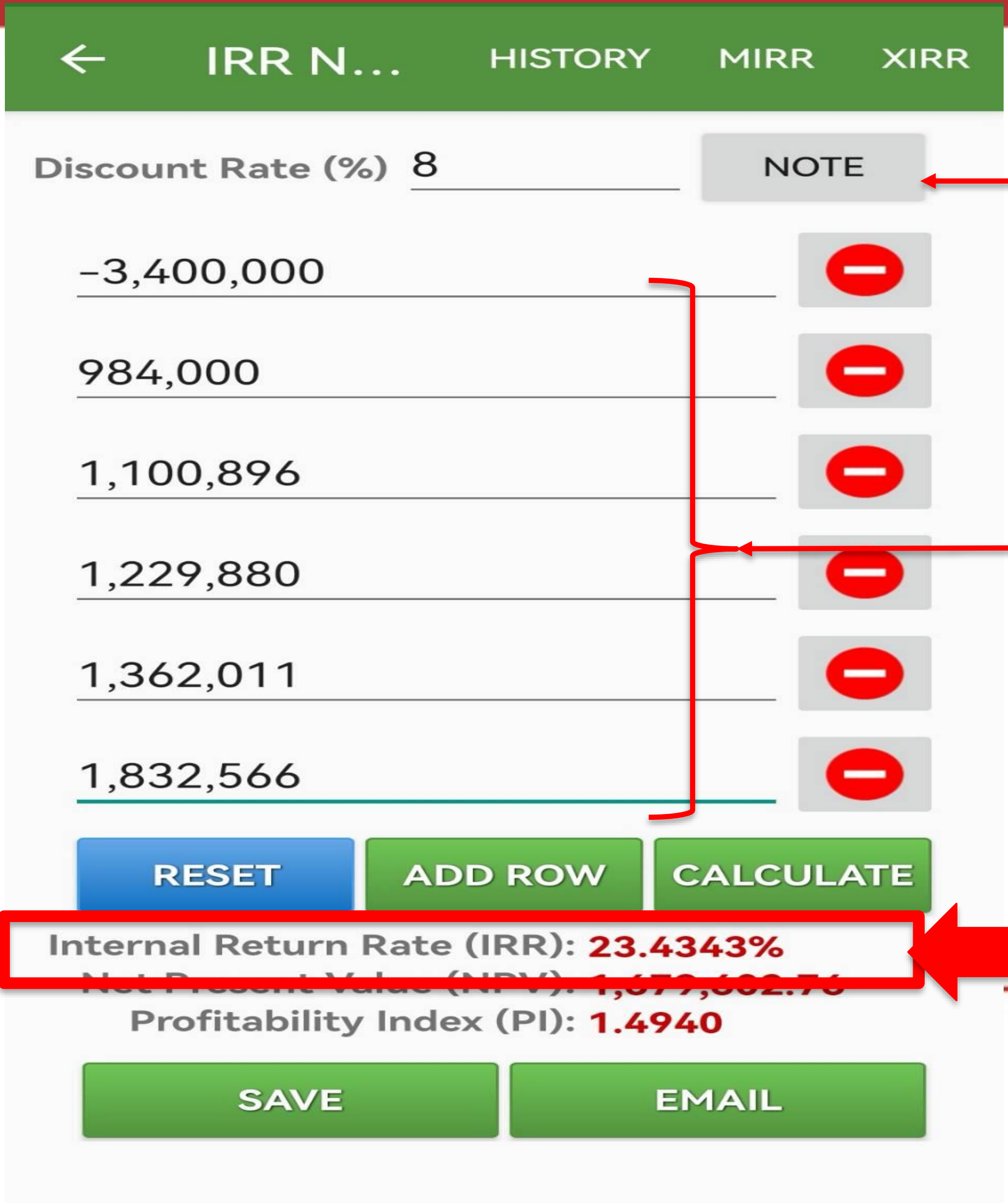
- The discount rate that forces the PV of the inflows to equal to an initial cost (or PV of all the costs if costs are incurred for several years). NPV = 0
- It uses to estimate **profitability** of the potential investment project.
- Rate of return that the owner will earn if implementing the project
- IRR can be found by the following equation:

$$\mathrm{NPV} = \mathrm{CF}_0 + \frac{CF_1}{(1+IRR)^1} + \frac{CF_2}{(1+IRR)^2} + \ldots + + \frac{CF_N}{(1+IRR)^N} = 0$$

- To **accept** the project, **IRR must higher than cost of capital (WACC)**

Note: The **higher IRR**, the **more desirable of** the project.

Internal Rate of Return (IRR): Calculator



- **Use IRR NPV calculator**
- Assume that an interest rate on loan (cost of capital) is **8%** per year.

Put cash flow that estimated earlier in these lines

Answer: IRR = 23.43%
Rate of return from this project is 23.34%
Accept the project since IRR more than cost of capital (8%).
The higher, the better.

Internal Rate of Return(IRR): Excel

For Excel, use **IRR** function

=IRR(value1, Value2, …….)

=IRR(B2:G2)

B4 =IRR(B2:G2)

	A	B	C	D	E	F	G
1	Year	0	1	2	3	4	5
2	Project's Cash Flow	(3,400,000.00)	984,000.00	1,100,896.00	1,229,880.00	1,362,010.00	1,832,565.00
3							
4	IRR	23.43%					
5							

Profitability Index (PI)

Profitability Index (PI)

Profitability Index (PI)

- The present value of future cash flow **divided** by the initial cost.
- The present value per dollar of the initial cost.
- **Profit compared to** a dollar that an entrepreneur's **investment**

$$PI = \frac{\text{PV of future cash flows}}{\text{Initial cost}} = \frac{\sum_{t=1}^{N} \frac{CF_t}{(1+r)^t}}{CF_0}$$

Where CF_0 represent the initial cost
CF_t represent the expected future cash flow

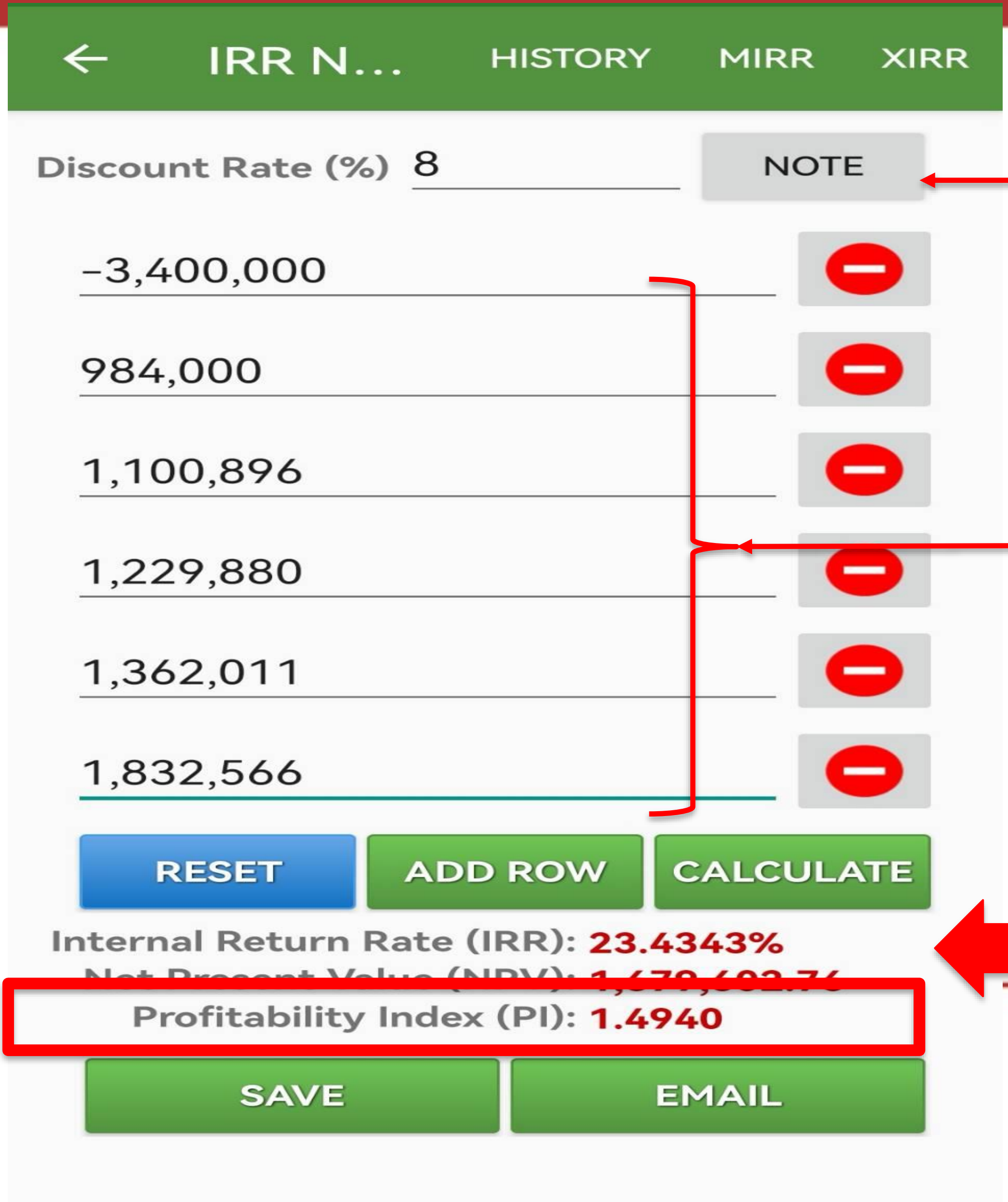
Profitability Index (PI)

Profitability Index (PI)- Present value per dollar of the initial cost.

$$PI = \frac{\text{PV of future cash flows}}{\text{Initial cost}} = \frac{\sum_{t=1}^{N} \frac{CF_t}{(1+r)^t}}{CF_0}$$

- **Accept** the project if PI is **greater than 1**
- **The higher PI**, the **higher** the project's ranking

Profitability Index (PI): Calculator



- **Use IRR NPV calculator**
- Assume that an interest rate on loan (cost of capital) is **8%** per year.

Put cash flow that estimated earlier in these lines

Answer: PI = 1.49
An entrepreneur's profitability is 1.49 times compared to their initial investment.
Accept the project since PI more than 1.
The higher, the better.

Profitability Index (PI): Excel

For Excel, use **NPV** function
=NPV(rate, value1, Value2, …….) / initial cash flow (year 0)
=NPV(8%,C2:G2) / B2

B4 | =NPV(0.08,C2:G2)/B2

	A	B	C	D	E	F	G
1	Year	0	1	2	3	4	5
2	Project's Cash Flow	3,400,000.00	984,000.00	1,100,896.00	1,229,880.00	1,362,010.00	1,832,565.00
3							
4	Profitability Index (PI)	$1.49					
5							

Note: Put positive number of initial cost

Payback Period

Payback Period

Payback Period

- The number of years required to recover the funds invested in a project from its operating cash flow.
- **The shorter the payback period, the better the project.**

Steps in Finding Payback Period

1. Find cumulative cash flow
2. Find a payback period by the following equation.

$$\text{Payback} = \text{Number of years prior to full recovery} + \frac{\text{Unrecovered cost at start of year}}{\text{Cash flow during full recovery year}}$$

Payback Period

Step 1: Find Cumulative Cash Flow

Year	0	1	2	3	4	5
Project's Cash Flow	-3,400,000	984,000	1,100,896	1,229,880	1,362,010	1,832,565
Cummulative CF	-3,400,000	-2,416,000	-1,315,104	-85,224	1,276,786	3,109,351

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Payback Period

Step 1: Find Cumulative Cash Flow

Year	0	1	2	3	4	5
Project's Cash Flow	-3,400,000	984,000	1,100,896	1,229,880	1,362,010	1,832,565
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Payback Period

Step 1: Find Cumulative Cash Flow

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Cummulative CF	-3,400,000	-2,416,000	-1,315,104	-85,224	1,276,786	3,109,351

Payback Period

Step 2: Put number into the formula

Year Prior to full recovery

Cash Flow during Full recovery year

Year	0	1	2	3	4	5
Project's Cash Flow	-3,400,000	984,000	1,100,896	1,229,880	1,362,010	1,832,565
Cummulative CF	-3,400,000	-2,416,000	-1,315,104	-85,224	1,276,786	3,109,351

Uncovered Cost

$$\text{Payback} = \text{Number of years prior to full recovery} + \frac{\text{Unrecovered cost at start of year}}{\text{Cash flow during full recovery year}}$$

$$\text{Payback} = 3 + \frac{85{,}224}{1{,}362{,}010} = 3.06$$

Project's Decision Criteria

Decision's Criteria (WACC = 8%)		Accept Project if
NPV	1,679,601.34	NPV is positive
IRR	23.43%	IRR > WACC
PI	1.49	PI > 1
Payback Period	3.06 years	

Q: Should SP Baker Co. Ltd. purchase and install this new machine?

A: Yes.

- Based on Net Present Value, **NPV** of SP Baker's project is **1,679,601** which indicates that this project would generate a cash flow to an entrepreneur of 1,679,601 baht.
- Internal Rate of Return (or **IRR**) of the project is **23.43%** indicating that the project would generate a rate of return of 23.43% which is higher than its cost of capital of 8%.
- Profitability index (**PI**) of the project is **1.49** which is more than 1 indicating that the project would be able to generate cash inflow more than its costs.
- Finally, it takes about 3 years to recover all funds invested in this project.

THE END.